

Majorization does not order the hazard rates of exponential mixtures with more than two components

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Abstract

For a finite mixture of exponential distributions with weights p and rates v , a known two-component fact is that spreading the rates by a T -transform lowers the hazard rate everywhere: if λ majorizes γ then the mixture with rates λ is the larger one in the hazard rate order. Theorem 6.3 of Shojaee, Asadi and Finkelstein (2022) proves such a statement for two-component generalized finite α -mixtures, and Remark 6.3 there asks whether it holds for more than two components; Theorem 3.8 in the accepted manuscript of Sahoo, Kayal and Finkelstein (2026) asserts an n -component version. We show that the n -component implication is false for every $n \geq 3$, already for equal weights and integer rates. The hazard rates of the equal-weight mixtures with rates $(1, 3, 5)$ and $(1, 4, 4)$ cross exactly once, at a time $t_* \in (\log 2, \log 4)$ that is the logarithm of an algebraic number of degree four, and adjoining any number of common rate-one components preserves the crossing. The mechanism is a common slowest component: for a general three-parameter family we prove that the hazard rates cross exactly once whenever the shared component is the slowest one, and we give a criterion (equal minimal rate, different second rate) that forces a crossing under majorization in every dimension. Variants with strictly ordered weights and rates, and with distinct minimal rates and two crossings, are certified with exact rational arithmetic. The usual stochastic order of the mixtures studied here, the ordering of the hazard rates near the origin and the two-component statement itself all survive. Consequently the extension asked for in Remark 6.3 of Shojaee, Asadi and Finkelstein has a negative answer, and Theorem 3.8, Corollary 3.2 and Theorem 3.9 of the accepted manuscript of Sahoo, Kayal and Finkelstein are false as stated. The same instances contradict the companion reversed hazard rate statements there (Theorem 3.11, Corollary 3.3 and Theorem 3.12) when the reversed hazard rate is the expression displayed in their proofs, and Theorem 3.11 also fails for the reversed hazard rate of the model as printed. A companion exact-arithmetic audit of the two manuscripts adds further certified failures: the complementary halves of Theorems 3.8 and 3.11 (the class $\mathcal{W}_n$ with $\alpha \leq 0$) fail as well, the stochastic-order statement Theorem 3.4 of Sahoo, Kayal and Finkelstein fails for $\alpha > 0$, and their Theorem 3.10 fails under the literal reading of the model; the audit also finds that two counterexamples printed in that manuscript (its Counterexamples 3.7 and 3.8) are themselves invalid (one reports a negative value that is in fact positive, the other a sign change that is floating-point noise), and it turns up minor errata in Shojaee, Asadi and Finkelstein (a boundary case of Theorem 4.1(b), a missing weight factor in Example 3.6, and a sign convention in Theorem 5.2(i) worth clarifying).

Keywords. finite mixture; hazard rate order; majorization; T -transform; exponential distribution; counterexample.

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1 Introduction

Let $p = (p_1, \dots, p_n)$ be a probability vector with positive entries and let $v = (v_1, \dots, v_n) \in (0, \infty)^n$. The finite (arithmetic) mixture of exponential distributions with weights p and rates v has survival function and hazard rate

$$S_{p,v}(t) = \sum_{i=1}^n p_i e^{-v_i t}, \quad h_{p,v}(t) = \frac{\sum_{i=1}^n p_i v_i e^{-v_i t}}{\sum_{i=1}^n p_i e^{-v_i t}}, \quad t \geq 0. \quad (1)$$

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When p is the uniform vector we write S_v and h_v . A vector λ majorizes γ , written $\lambda \succ \gamma$, when λ is more spread out than γ with the same total (Definition 2.1). The question studied here is the following.

If $\lambda \succ \gamma$, is $h_{p,\lambda} \leq h_{p,\gamma}$ on $[0, \infty)$, that is, is the mixture with the more heterogeneous rates the larger one in the hazard rate order?

For two components and equal weights the answer is yes: with $\lambda = (b - d, b + d)$ and $\gamma = (b, b)$ one has $S_\lambda(t) = e^{-bt} \cosh(dt)$, hence $h_\lambda(t) = b - d \tanh(dt) \leq b = h_\gamma(t)$. Theorem 6.3 of Shojaei, Asadi and Finkelstein [16] proves a statement of this type for two-component generalized finite α -mixtures with common mixing proportions, under an antiordering condition between weights and parameters; ordinary exponential mixtures are the special case $\alpha_i = 1$ with baseline $e^{-\lambda t}$, which is exactly the baseline of their Example 6.4. Remark 6.3 of the same paper reads: “Theorem 6.3 is proved for $n = 2$. We have not yet been able to prove whether the result holds for $n > 2$ or not leaving it as an open problem for now”, with a pointer to [8, 11] for the same question about ordinary mixtures. Theorem 3.8 in the accepted manuscript of Sahoo, Kayal and Finkelstein [14] asserts an n -component version for finite α -mixtures with resilience-scaled components: if the $2 \times n$ matrix formed by the weights and the resilience parameters is multiplied by a single T -transform, the original mixture is the larger one in the hazard rate order.

Answer. The n -component implication is false for every $n \geq 3$, and it fails in the simplest possible setting. Our results are the following.

- (a) **Three components** (Theorem 3.1). For equal weights, $\lambda = (1, 3, 5) \succ \gamma = (1, 4, 4)$, and with $q = e^{-t}$,

$$h_\lambda(t) - h_\gamma(t) = \frac{2q^2(1-q)(1-2q-q^3-q^4)}{(1+q^2+q^4)(1+2q^3)}.$$

The difference is negative on $(0, t_*)$ and positive on (t_*, ∞) , where $t_* = -\log q_* \approx 0.8231$ and $q_* \approx 0.4391$ is the unique root of $1 - 2q - q^3 - q^4$ in $(0, 1)$. In particular $h_\lambda(\log 2) - h_\gamma(\log 2) = -1/35$ and $h_\lambda(\log 4) - h_\gamma(\log 4) = 41/1001$. Neither mixture dominates the other in the hazard rate order.

- (b) **Every dimension** (Theorem 4.1). Adjoining m common rate-one components gives $n = m + 2$ component equal-weight mixtures with rates $(1, \dots, 1, 3, 5) \succ (1, \dots, 1, 4, 4)$ whose hazard rates cross exactly once, at a time $t_*(m) \in (\log 2, \log 4)$ that decreases strictly to $\log 2$; the difference at $t = \log 4$ equals $3(128m - 5)/((256m + 17)(32m + 1)) > 0$ for every $m \geq 1$.
- (c) **Mechanism** (Theorems 5.1 and 5.4). In the family with weights $(w, \frac{1-w}{2}, \frac{1-w}{2})$ and rates $(a, b - d, b + d)$ against (a, b, b) the hazard rates cross exactly once when $w \in (0, 1)$ and $a < b - d$, and the hazard rate order holds in all other cases. More generally, if two mixtures have the same minimal rate, the mixture with the smaller second rate has the larger hazard rate for large t , while for equal weights majorization forces the opposite inequality near $t = 0$; this gives a crossing for every n .
- (d) **Variants** (Propositions 6.1 and 6.2). The failure persists with strictly decreasing weights and strictly increasing rates (so all antiordering conditions hold strictly), and it persists when the minimal rates differ, where the sign of the hazard difference changes twice.
- (e) **Relation to the published statements** (Section 7). The extension asked for in Remark 6.3 of [16] has a negative answer for every $n \geq 3$. Theorem 3.8 of the accepted manuscript of [14], and with it Corollary 3.2 and Theorem 3.9 there, are contradicted by explicit instances; we point to the step of the published proof that does not apply. The same instances contradict the reversed hazard rate statements Theorem 3.11, Corollary 3.3 and Theorem 3.12 there (Section 7.3). A companion exact-arithmetic audit of both manuscripts (Section 9) certifies further failures in [14]: the complementary $\mathcal{W}_n$, $\alpha \leq 0$ halves of Theorems 3.8 and 3.11 (so both halves of both statements fail), the stochastic-order statement Theorem 3.4 for $\alpha > 0$, and Theorem 3.10 under the literal reading of the model; it also shows that the counterexamples printed there as Counterexamples 3.7 and 3.8 are themselves invalid, and records minor errata in [16]. The usual stochastic order of the mixtures studied here (Theorem 6.1 of [16]), the two-component statements (Theorem 6.3 there and Theorem 3.7 of [14], and Theorem 3.10 under the substantive reading), and the ordering of the hazard rates on a neighbourhood of the origin are not affected.

Why it matters. Hazard-rate comparisons of mixtures are the standard tool for comparing heterogeneous populations in reliability, and majorization is the standard measure of heterogeneity of the component parameters. The two-component intuition (more heterogeneity, smaller hazard) is correct near the origin in every dimension, because $h'_{p,v}(0) = -\text{Var}_p(v)$ and the variance is Schur-convex. It is wrong in the tail, where $h_{p,v}(t)$ converges to the minimal rate at a speed governed by the second smallest rate, and majorization does not control that rate in the required direction. The failure is therefore structural, not an artefact of a particular parametrization: no proof strategy can establish the n -component statement, and only weaker conclusions (Proposition 7.3) remain.

Novelty boundary. We claim no new positive ordering result. The facts in Lemma 2.2 are classical (the decreasing hazard rate of exponential mixtures goes back to Proschan [12]), and the two-component hazard rate statements of [14, 16] are not contradicted. A bounded literature search (Section 8) found no earlier counterexample to the n -component statement and no correction of [14]; we cannot exclude that the failure has been noticed before. All crossing counts are established by hand (Theorems 3.1, 4.1, 5.1), by Descartes' rule of signs applied to exactly computed polynomials (Proposition 6.1), or by exact computer algebra (Proposition 6.2, where the existence of the crossings is proved by hand and the exact count is machine-certified).

2 Preliminaries

Definition 2.1 (majorization). For $x \in \mathbb{R}^n$ let $x_{[1]} \geq \dots \geq x_{[n]}$ be its decreasing rearrangement. The vector x majorizes y , written $x \succ y$, if $\sum_{i=1}^k x_{[i]} \geq \sum_{i=1}^k y_{[i]}$ for $k = 1, \dots, n-1$ and $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$ [9, Chapter 1]. Equivalently, with the increasing rearrangements $x_{(1)} \leq \dots \leq x_{(n)}$, $\sum_{i \leq k} x_{(i)} \leq \sum_{i \leq k} y_{(i)}$ for $k < n$ with equality for $k = n$; this is Definition 6.1 of [16].

A T -transform is a matrix $T = \omega I + (1 - \omega)\Pi$ with $\omega \in [0, 1]$ and Π a permutation matrix that interchanges two coordinates [9, Chapter 2.B]. Acting on a row vector x from the right, T replaces a pair (x_i, x_j) by $(\omega x_i + (1 - \omega)x_j, (1 - \omega)x_i + \omega x_j)$ and leaves the other coordinates unchanged; acting on a $2 \times n$ matrix it does this to both rows. Always $x \succ xT$. Conversely, if $x \succ y$ then $y = xT_1 \dots T_k$ for finitely many T -transforms [9, Lemma 2.B.1].

Let X and Y be nonnegative random variables with survival functions S_X, S_Y and hazard rates h_X, h_Y . Then $X \leq_{\text{st}} Y$ if $S_X \leq S_Y$ on $[0, \infty)$, and $X \leq_{\text{hr}} Y$ if $h_X(t) \geq h_Y(t)$ for all $t \geq 0$ [15, Sections 1.A and 1.B]; the hazard rate order implies the usual stochastic order. Both [16] and [14] use the same convention (Definition 2.1(ii) of the latter). Thus “the mixture with rates λ is larger in the hazard rate order” means $h_{p,\lambda} \leq h_{p,\gamma}$. If X, Y have distribution functions F_X, F_Y and densities f_X, f_Y , the reversed hazard rates are $r_X = f_X/F_X$ and $r_Y = f_Y/F_Y$, and $X \leq_{\text{rh}} Y$ if $r_X(t) \leq r_Y(t)$ for all $t > 0$ (Definition 2.1(iii) of [14]); the reversed hazard rate order also implies the usual stochastic order [15, Section 1.B].

We use the substitution $q = e^{-t} \in (0, 1)$ throughout. If the rates are integers, then by (1) both $S_{p,v}$ and $h_{p,v}$ become rational functions of q with positive denominators on $(0, 1)$; if the rates are rational with common denominator N , the same holds with $s = e^{-t/N}$. Small t corresponds to q near 1 and large t to q near 0.

Lemma 2.2 (basic facts). Let p have positive entries and let $v \in (0, \infty)^n$ be non-constant. Write $\mu = \sum_i p_i v_i$, $\sigma^2 = \sum_i p_i (v_i - \mu)^2$, $v_{\min} = \min_i v_i$, $W_1 = \sum_{i: v_i = v_{\min}} p_i$, $u = \min\{v_i : v_i > v_{\min}\}$ and $W_2 = \sum_{i: v_i = u} p_i$. Then:

- (a) $h_{p,v}$ is real-analytic on $[0, \infty)$, $h_{p,v}(0) = \mu$ and $h'_{p,v}(0) = -\sigma^2$;
- (b) $h_{p,v}$ is strictly decreasing on $[0, \infty)$;
- (c) $h_{p,v}(t) \rightarrow v_{\min}$ as $t \rightarrow \infty$, and more precisely

$$h_{p,v}(t) - v_{\min} = \frac{W_2}{W_1} (u - v_{\min}) e^{-(u - v_{\min})t} (1 + o(1)), \quad t \rightarrow \infty.$$

Proof. Write $S = S_{p,v}$ and $N = -S' = \sum_i p_i v_i e^{-v_i t}$, so that $h = N/S$ and $S > 0$; this gives analyticity. Since $N' = -\sum_i p_i v_i^2 e^{-v_i t}$,

$$h' = \frac{N'S - NS'}{S^2} = \frac{N'S + N^2}{S^2} = -\left(\sum_i \pi_i(t) v_i^2 - \left(\sum_i \pi_i(t) v_i\right)^2\right) = -\text{Var}_{\pi(t)}(v),$$

where $\pi_i(t) = p_i e^{-v_i t} / S(t)$ is a probability vector with positive entries. The variance of the non-constant vector v under a probability vector with positive entries is strictly positive, which proves (b); at $t = 0$ we have $\pi(0) = p$, which proves (a). For (c) write

$$h_{p,v}(t) - v_{\min} = \frac{\sum_i p_i (v_i - v_{\min}) e^{-(v_i - v_{\min})t}}{\sum_i p_i e^{-(v_i - v_{\min})t}}.$$

The denominator tends to W_1 . After multiplication by $e^{(u - v_{\min})t}$ the numerator equals $W_2(u - v_{\min}) + \sum_{i: v_i > u} p_i (v_i - v_{\min}) e^{-(v_i - u)t}$, which tends to $W_2(u - v_{\min})$. $\square$

Lemma 2.3 (*T-transforms and convex sums*). *Let $x \succ y$ in $\mathbb{R}^n$.*

(a) *For every convex $\varphi : \mathbb{R} \rightarrow \mathbb{R}$, $\sum_i \varphi(y_i) \leq \sum_i \varphi(x_i)$. If φ is strictly convex and y is not a rearrangement of x , the inequality is strict.*

(b) $\sum_i y_i^2 \leq \sum_i x_i^2$, *with equality if and only if y is a rearrangement of x .*

Proof. By [9, Lemma 2.B.1], $y = x T_1 \cdots T_k$ with T -transforms T_j . A single T -transform with parameter ω acting on the coordinates $(z_i, z_j) = (\alpha, \beta)$ of a vector z changes $\sum_l \varphi(z_l)$ by

$$\varphi(\omega\alpha + (1 - \omega)\beta) + \varphi((1 - \omega)\alpha + \omega\beta) - \varphi(\alpha) - \varphi(\beta) \leq 0,$$

by convexity applied to each of the two terms; for strictly convex φ the change is strictly negative unless $\omega \in \{0, 1\}$ or $\alpha = \beta$, and in those two cases the transform permutes the coordinates of z . Hence $\sum_l \varphi(z_l)$ is non-increasing along the chain, and if it is constant along the chain for a strictly convex φ then every step is a permutation and y is a rearrangement of x . This proves (a). For (b) take $\varphi(z) = z^2$, for which the change equals $-2\omega(1 - \omega)(\alpha - \beta)^2$. $\square$

3 Three components

Theorem 3.1. *Let $\lambda = (1, 3, 5)$ and $\gamma = (1, 4, 4)$ with equal weights $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and put $q = e^{-t}$.*

(a) $\lambda \succ \gamma$; indeed $\gamma = \lambda T$ with $T = \frac{1}{2}(I + \Pi_{23})$, where Π_{23} interchanges the last two coordinates.

(b) For $t > 0$,

$$h_\lambda(t) = \frac{1 + 3q^2 + 5q^4}{1 + q^2 + q^4}, \quad h_\gamma(t) = \frac{1 + 8q^3}{1 + 2q^3}, \quad (2)$$

and, with $B_1(q) = 1 - 2q - q^3 - q^4$,

$$h_\lambda(t) - h_\gamma(t) = \frac{2q^2(1 - q) B_1(q)}{(1 + q^2 + q^4)(1 + 2q^3)}. \quad (3)$$

(c) B_1 has exactly one root q_* in $(0, 1)$, and $\frac{1}{4} < q_* < \frac{1}{2}$. With $t_* = -\log q_* \in (\log 2, \log 4)$ one has $h_\lambda < h_\gamma$ on $(0, t_*)$, $h_\lambda(t_*) = h_\gamma(t_*)$ and $h_\lambda > h_\gamma$ on (t_*, ∞) . Numerically $q_* = 0.439087115146\dots$ and $t_* = 0.82305744562489\dots$ (both truncated, not rounded).

(d) $h_\lambda(\log 2) = \frac{11}{7}$, $h_\gamma(\log 2) = \frac{8}{5}$, $h_\lambda(\log 4) = \frac{103}{91}$, $h_\gamma(\log 4) = \frac{12}{11}$; hence

$$h_\lambda(\log 2) - h_\gamma(\log 2) = -\frac{1}{35} < 0, \quad h_\lambda(\log 4) - h_\gamma(\log 4) = \frac{41}{1001} > 0.$$

(e) $S_\lambda(t) - S_\gamma(t) = \frac{1}{3}q^3(1 - q)^2 > 0$ for $t > 0$; both mixtures have $h(0) = 3$, $h'_\lambda(0) = -\frac{8}{3} < -2 = h'_\gamma(0)$, and $h_\lambda(t), h_\gamma(t) \rightarrow 1$ as $t \rightarrow \infty$.

In particular the two mixtures are ordered in the usual stochastic order but in neither direction of the hazard rate order.

Proof. (a) The decreasing rearrangements are $(5, 3, 1)$ and $(4, 4, 1)$ with partial sums $5 \geq 4$, $8 \geq 8$, $9 = 9$. The T -transform $\frac{1}{2}(I + \Pi_{23})$ replaces $(3, 5)$ by $(4, 4)$.

(b) By (1) with equal weights, $h_\lambda = (q + 3q^3 + 5q^5)/(q + q^3 + q^5)$ and $h_\gamma = (q + 8q^4)/(q + 2q^4)$; cancel q . Expanding the cross products,

$$\begin{aligned} (1 + 3q^2 + 5q^4)(1 + 2q^3) - (1 + 8q^3)(1 + q^2 + q^4) &= 2q^2 - 6q^3 + 4q^4 - 2q^5 + 2q^7 \\ &= 2q^2(1 - q)(1 - 2q - q^3 - q^4), \end{aligned}$$

since $(1 - q)(1 - 2q - q^3 - q^4) = 1 - 3q + 2q^2 - q^3 + q^5$.

(c) $B'_1(q) = -2 - 3q^2 - 4q^3 < 0$, so B_1 is strictly decreasing on $[0, 1]$; $B_1(0) = 1$, $B_1(\frac{1}{4}) = \frac{1}{2} - \frac{1}{64} - \frac{1}{256} = \frac{123}{256} > 0$ and $B_1(\frac{1}{2}) = -\frac{3}{16} < 0$. Hence there is exactly one root $q_* \in (\frac{1}{4}, \frac{1}{2})$, $B_1 < 0$ on $(q_*, 1)$ and $B_1 > 0$ on $(0, q_*)$. Since $q \mapsto -\log q$ is decreasing, $q \in (q_*, 1)$ corresponds to $t \in (0, t_*)$. The other factors in (3) are positive for $q \in (0, 1)$. The numerical value is obtained by exact root isolation (Section 9); the reader can check $B_1(0.4390) > 0 > B_1(0.4392)$.

(d) Put $q = \frac{1}{2}$: $h_\lambda = \frac{33/16}{21/16} = \frac{11}{7}$ and $h_\gamma = \frac{2}{5/4} = \frac{8}{5}$. Put $q = \frac{1}{4}$: $h_\lambda = \frac{256+48+5}{256+16+1} = \frac{309}{273} = \frac{103}{91}$ and $h_\gamma = \frac{1+1/8}{1+1/32} = \frac{36}{33} = \frac{12}{11}$.

(e) $S_\lambda - S_\gamma = \frac{1}{3}(q^3 + q^5 - 2q^4) = \frac{1}{3}q^3(1 - q)^2$. The values at 0 and the derivatives follow from Lemma 2.2(a): both means equal 3, and $\text{Var}(1, 3, 5) = \frac{8}{3}$, $\text{Var}(1, 4, 4) = 2$. The limits follow from Lemma 2.2(c). $\square$

Remark 3.2. Theorem 3.1 is fully hand-checkable: the identity (3) is a polynomial identity of degree seven, and the sign change is witnessed by the two rational values in (d). The point t_* is the logarithm of an algebraic number of degree four, since B_1 is irreducible over $\mathbb{Q}$. Indeed $-B_1 = q^4 + q^3 + 2q - 1$ is monic with integer coefficients and reduces modulo 2 to $q^4 + q^3 + 1$, which has no root in $\mathbb{F}_2$ and is not divisible by $q^2 + q + 1$, the only irreducible quadratic over $\mathbb{F}_2$ (the remainder is q). So $q^4 + q^3 + 1$ is irreducible over $\mathbb{F}_2$, hence $-B_1$ is irreducible over $\mathbb{Z}$ and, by Gauss's lemma, over $\mathbb{Q}$. The verification script checks both this argument and irreducibility over $\mathbb{Q}$ directly.

4 Every number of components

For real $m > 0$ consider the survival functions

$$S_\lambda^{(m)}(t) = \frac{me^{-t} + e^{-3t} + e^{-5t}}{m + 2}, \quad S_\gamma^{(m)}(t) = \frac{me^{-t} + 2e^{-4t}}{m + 2}. \quad (4)$$

For a positive integer m these are the equal-weight mixtures with $n = m + 2$ components and rate vectors

$$\lambda^{(m)} = (\underbrace{1, \dots, 1}_m, 3, 5), \quad \gamma^{(m)} = (\underbrace{1, \dots, 1}_m, 4, 4),$$

and $\gamma^{(m)} = \lambda^{(m)}T$ with $T = \frac{1}{2}(I + \Pi_{n-1, n})$, so $\lambda^{(m)} \succ \gamma^{(m)}$. Let $h_\lambda^{(m)}$ and $h_\gamma^{(m)}$ denote the hazard rates and put

$$B_m(q) = m(1 - 2q) - q^3(1 + q). \quad (5)$$

Theorem 4.1. *Let $m > 0$ and $q = e^{-t}$.*

(a) *For $t > 0$,*

$$h_\lambda^{(m)}(t) - h_\gamma^{(m)}(t) = \frac{2q^2(1 - q)B_m(q)}{(m + q^2 + q^4)(m + 2q^3)}. \quad (6)$$

(b) *B_m has exactly one root $q_*(m)$ in $(0, 1)$, and $0 < q_*(m) < \frac{1}{2}$. With $t_*(m) = -\log q_*(m)$ one has $h_\lambda^{(m)} < h_\gamma^{(m)}$ on $(0, t_*(m))$ and $h_\lambda^{(m)} > h_\gamma^{(m)}$ on $(t_*(m), \infty)$.*

(c) *For $m \geq 1$: $\frac{1}{2} - \frac{3}{16m} < q_*(m) < \frac{1}{2}$, hence $\log 2 < t_*(m) < \log 2 + \frac{3}{8m-3} < \log 4$. Moreover*

$$\begin{aligned} h_\lambda^{(m)}(\log 2) - h_\gamma^{(m)}(\log 2) &= -\frac{3}{(16m + 5)(4m + 1)} < 0, \\ h_\lambda^{(m)}(\log 4) - h_\gamma^{(m)}(\log 4) &= \frac{3(128m - 5)}{(256m + 17)(32m + 1)} > 0. \end{aligned}$$

Table 1: Crossing times of the padded family (Theorem 4.1), obtained by exact root isolation and rounded to six decimals; $\log 2 \approx 0.693147$.

$n = m + 2$	$q_*(m)$	$t_*(m)$
3	0.439087	0.823057
4	0.463554	0.768832
5	0.473863	0.746838
6	0.479597	0.734808
8	0.485804	0.721950
66	0.498549	0.696053

(d) $m \mapsto q_*(m)$ is strictly increasing on $(0, \infty)$, so $t_*(m)$ decreases strictly to $\log 2$ as $m \rightarrow \infty$.

(e) $S_\lambda^{(m)}(t) - S_\gamma^{(m)}(t) = q^3(1 - q)^2/(m + 2) > 0$ for $t > 0$.

Consequently, for every $n \geq 3$ there are equal-weight exponential mixtures with n components whose integer rate vectors satisfy $\lambda \succ \gamma$ (with $\gamma = \lambda T$ for a single T -transform) and whose hazard rates cross exactly once; in particular they are not comparable in the hazard rate order.

Proof. (a) From (4), $h_\lambda^{(m)} = (m + 3q^2 + 5q^4)/(m + q^2 + q^4)$ and $h_\gamma^{(m)} = (m + 8q^3)/(m + 2q^3)$. The numerator of the difference is

$$(m + 3q^2 + 5q^4)(m + 2q^3) - (m + 8q^3)(m + q^2 + q^4) = 2mq^2 - 6mq^3 + 4mq^4 - 2q^5 + 2q^7,$$

and the right-hand side equals $2q^2[m(1 - q)(1 - 2q) - q^3(1 - q^2)] = 2q^2(1 - q)B_m(q)$.

(b) $B'_m(q) = -2m - 3q^2 - 4q^3 < 0$ on $[0, 1]$, $B_m(0) = m > 0$ and $B_m(\frac{1}{2}) = -\frac{3}{16} < 0$. The sign pattern follows as in Theorem 3.1(c) because the remaining factors of (6) are positive on $(0, 1)$.

(c) Let $q_0 = \frac{1}{2} - \frac{3}{16m}$; for $m \geq 1$, $q_0 \in [\frac{5}{16}, \frac{1}{2})$. Then $m(1 - 2q_0) = \frac{3}{8}$ and $q_0^3(1 + q_0) < (\frac{1}{2})^3 \cdot \frac{3}{2} = \frac{3}{16}$, so $B_m(q_0) > \frac{3}{16} > 0$ and therefore $q_0 < q_*(m)$ by monotonicity of B_m . Since $-\log(1 - x) \leq x/(1 - x)$ for $x \in [0, 1]$, $t_*(m) < -\log q_0 = \log 2 - \log(1 - \frac{3}{8m}) \leq \log 2 + \frac{3}{8m-3}$, and $\frac{3}{8m-3} \leq \frac{3}{5} < \log 2$. The two values follow from (6) with $q = \frac{1}{2}$, where $B_m = -\frac{3}{16}$, and $q = \frac{1}{4}$, where $B_m = \frac{m}{2} - \frac{5}{256}$; for $m = 1$ they reduce to $-\frac{1}{35}$ and $\frac{41}{1001}$.

(d) For $m' > m$ and $q < \frac{1}{2}$, $B_{m'}(q) - B_m(q) = (m' - m)(1 - 2q) > 0$. Hence $B_{m'}(q_*(m)) > B_m(q_*(m)) = 0 = B_{m'}(q_*(m'))$, and since $B_{m'}$ is strictly decreasing, $q_*(m') > q_*(m)$. The limit follows from (c).

(e) $S_\lambda^{(m)} - S_\gamma^{(m)} = (q^3 + q^5 - 2q^4)/(m + 2)$. □

Remark 4.2 (the two-component case). Formula (6) remains valid for $m = 0$, where $B_0(q) = -q^3(1 + q) < 0$ and the difference reduces to $-(1 - q^2)/(1 + q^2) = -\tanh t$: the hazard rate order $h_{(3,5)} \leq h_{(4,4)}$ holds, in agreement with the two-component statements. The crossing appears exactly when a component that is slower than both transformed components is present.

5 The mechanism

5.1 A common slow component

Fix $a > 0$, $b > d > 0$ and $w \in [0, 1]$, and let X and Y be the mixtures with common weights $(w, \frac{1-w}{2}, \frac{1-w}{2})$ and rates $(a, b - d, b + d)$ and (a, b, b) respectively, so that the rate vector of X majorizes that of Y (they are related by $T = \frac{1}{2}(I + \Pi_{23})$). Their survival functions are

$$S_X(t) = we^{-at} + (1 - w)e^{-bt} \cosh(dt), \quad S_Y(t) = we^{-at} + (1 - w)e^{-bt}. \quad (7)$$

Put $c = b - a$.

Theorem 5.1. (a) $S_X(t) > S_Y(t)$ for all $t > 0$; in particular $X \succeq_{\text{st}} Y$.

(b) If $w \in (0, 1)$ and $a < b - d$, there is a unique $t_* > 0$ such that $h_X < h_Y$ on $(0, t_*)$, $h_X(t_*) = h_Y(t_*)$ and $h_X > h_Y$ on (t_*, ∞) . The time t_* is the unique solution of

$$d \coth\left(\frac{dt}{2}\right) = \frac{cw e^{ct}}{w e^{ct} + 1 - w}. \quad (8)$$

(c) If $w = 0$, or if $w \in (0, 1)$ and $a \geq b - d$, then $h_X(t) < h_Y(t)$ for all $t > 0$, so $X \geq_{\text{hr}} Y$. For $w = 0$ one has $h_X(t) = b - d \tanh(dt)$ and $h_Y \equiv b$.

Proof. (a) $S_X - S_Y = (1 - w)e^{-bt}(\cosh(dt) - 1)$ and $\cosh(dt) - 1 = 2 \sinh^2(dt/2) > 0$ for $t > 0$.

Let $R = S_X/S_Y$. Since $h_X = -(\log S_X)'$ and $h_Y = -(\log S_Y)'$,

$$h_X - h_Y = -(\log R)' = -\frac{R'}{R}, \quad \text{so} \quad \text{sign}(h_X - h_Y) = -\text{sign}(R'). \quad (9)$$

Writing $S_Y = e^{-bt}(w e^{ct} + 1 - w)$ we obtain

$$R(t) - 1 = \frac{(1 - w)(\cosh(dt) - 1)}{w e^{ct} + 1 - w} > 0 \quad (t > 0),$$

and therefore $\text{sign}(R') = \text{sign}((\log(R - 1))') = \text{sign } F(t)$ with

$$F(t) = \frac{d \sinh(dt)}{\cosh(dt) - 1} - \frac{cw e^{ct}}{w e^{ct} + 1 - w} = d \coth\left(\frac{dt}{2}\right) - G(t), \quad G(t) = \frac{cw e^{ct}}{w e^{ct} + 1 - w}.$$

(c) If $w = 0$ then $G \equiv 0$ and $F = d \coth(dt/2) > 0$ on $(0, \infty)$. If $w \in (0, 1)$ and $c \leq 0$ then $G \leq 0$ and again $F > 0$. If $w \in (0, 1)$ and $0 < c \leq d$ then $0 < G(t) < c \leq d < d \coth(dt/2)$, so $F > 0$. In all three cases R is strictly increasing on $(0, \infty)$ and (9) gives $h_X < h_Y$ for $t > 0$. The condition $a \geq b - d$ is $c \leq d$. For $w = 0$, $S_X = e^{-bt} \cosh(dt)$ gives $h_X = b - d \tanh(dt)$ directly.

(b) Now $w \in (0, 1)$ and $c > d > 0$. The function $t \mapsto d \coth(dt/2)$ is continuous and strictly decreasing on $(0, \infty)$ (its derivative is $-d^2/(2 \sinh^2(dt/2))$), with limits $+\infty$ at 0^+ and d at ∞ . The function G is continuous and strictly increasing (its derivative is $c^2 w(1 - w)e^{ct}/(w e^{ct} + 1 - w)^2 > 0$), with limits cw at 0^+ and c at ∞ . Hence F is continuous and strictly decreasing on $(0, \infty)$, with $F(0^+) = +\infty$ and $F(\infty) = d - c < 0$. So F has a unique zero t_* , is positive before it and negative after it. By (9), $h_X - h_Y$ is negative on $(0, t_*)$, zero at t_* and positive on (t_*, ∞) . Equation (8) is $F(t) = 0$. $\square$

Corollary 5.2. With $(a, b, d) = (1, 4, 1)$ and $w = m/(m + 2)$, $m > 0$, the pair (X, Y) of Theorem 5.1 is the padded family (4), and in the variable $q = e^{-t}$

$$F(t) = \frac{1 + q}{1 - q} - \frac{3m}{m + 2q^3} = -\frac{2B_m(q)}{(1 - q)(m + 2q^3)}.$$

The weights are antiorordered with the rates, $(p_i - p_j)(v_i - v_j) \leq 0$ for all i, j , exactly when $w \geq \frac{1}{3}$, that is $m \geq 1$; the crossing occurs for every $m > 0$.

Proof. $w(m + 2) = m$ and $(1 - w)(m + 2) = 2$ give (4). Here $c = 3$, $\coth(t/2) = (1 + q)/(1 - q)$ and $e^{3t} = q^{-3}$, so $G = 3wq^{-3}/(wq^{-3} + 1 - w) = 3m/(m + 2q^3)$, and $(1 + q)(m + 2q^3) - 3m(1 - q) = -2m + 4mq + 2q^3 + 2q^4 = -2B_m(q)$. The antiorordering condition for the weights $(w, \frac{1-w}{2}, \frac{1-w}{2})$ and the rates $(1, 3, 5)$ or $(1, 4, 4)$ is $w \geq \frac{1-w}{2}$. $\square$

Remark 5.3 (majorization alone decides nothing). The boundary case $a = b - d$ of Theorem 5.1(c) with $(a, b, d, w) = (1, 4, 3, \frac{1}{3})$ is the pair of equal-weight mixtures with rates $(1, 1, 7)$ and $(1, 4, 4)$. Here $(1, 1, 7) \succ (1, 4, 4)$ and the hazard rate order $h_{(1,1,7)} < h_{(1,4,4)}$ holds on $(0, \infty)$; explicitly

$$h_{(1,1,7)}(t) - h_{(1,4,4)}(t) = \frac{6q^3(q - 1)(q^3 + 2)(q^2 + q + 1)}{(2q^3 + 1)(q^6 + 2)}.$$

Thus among equal-weight three-component mixtures with $\lambda \succ \gamma$ the hazard rate order sometimes holds and sometimes fails; the deciding quantity is the tail behaviour described next.

5.2 A crossing criterion

Theorem 5.4. *Let p have positive entries and let $\lambda, \gamma \in (0, \infty)^n$ be non-constant, with p -means μ_λ, μ_γ and p -variances $\sigma_\lambda^2, \sigma_\gamma^2$ as in Lemma 2.2. Suppose*

- (A) *either $\mu_\lambda < \mu_\gamma$, or $\mu_\lambda = \mu_\gamma$ and $\sigma_\lambda^2 > \sigma_\gamma^2$; and*
- (B) *$\min_i \lambda_i = \min_i \gamma_i$, and the second smallest distinct value of λ is smaller than the second smallest distinct value of γ .*

Then there are $0 < \varepsilon < T < \infty$ with $h_{p,\lambda} < h_{p,\gamma}$ on $(0, \varepsilon]$ and $h_{p,\lambda} > h_{p,\gamma}$ on $[T, \infty)$. In particular the two mixtures are not comparable in the hazard rate order.

Proof. By Lemma 2.2(a), $h_{p,\lambda}(0) = \mu_\lambda$, $h_{p,\gamma}(0) = \mu_\gamma$, $h'_{p,\lambda}(0) = -\sigma_\lambda^2$ and $h'_{p,\gamma}(0) = -\sigma_\gamma^2$; under (A) the difference $h_{p,\lambda} - h_{p,\gamma}$ is negative at 0, or vanishes at 0 with negative derivative, and in both cases it is negative on some $(0, \varepsilon]$. Let v_1 be the common minimum and $u_\lambda < u_\gamma$ the second smallest distinct values. By Lemma 2.2(c),

$$e^{(u_\lambda - v_1)t} (h_{p,\lambda}(t) - v_1) \rightarrow \frac{W_2^\lambda}{W_1^\lambda} (u_\lambda - v_1) > 0, \quad e^{(u_\lambda - v_1)t} (h_{p,\gamma}(t) - v_1) = e^{-(u_\gamma - u_\lambda)t} O(1) \rightarrow 0,$$

so $h_{p,\lambda} - h_{p,\gamma} > 0$ for all large t . □

Corollary 5.5. *Let p be uniform and $\lambda \succ \gamma$, where γ is not a rearrangement of λ . Suppose $\min_i \lambda_i = \min_i \gamma_i =: v_1$, that v_1 has the same multiplicity k in both vectors, and that $\lambda_{(k+1)} < \gamma_{(k+1)}$. Then the hazard rates of the two equal-weight mixtures cross: neither hazard rate order holds.*

Proof. Majorization gives $\mu_\lambda = \mu_\gamma$, and Lemma 2.3(b) gives $\sigma_\lambda^2 > \sigma_\gamma^2$, so (A) holds. Equal multiplicity of the minimum means that the second smallest distinct values are $\lambda_{(k+1)}$ and $\gamma_{(k+1)}$, so (B) holds. □

Remark 5.6. If $\lambda \succ \gamma$ and the minima coincide, then $\lambda_{(1)} = \gamma_{(1)}$ forces the multiplicity of the minimum in λ to be at least that in γ , and when the multiplicities are equal to k the partial sums give $\lambda_{(k+1)} \leq \gamma_{(k+1)}$. So Corollary 5.5 covers every equal-weight pair with a common minimum of equal multiplicity except the degenerate case $\lambda_{(k+1)} = \gamma_{(k+1)}$. The remaining configurations are a smaller minimum in λ (Proposition 6.2 shows that a crossing may still occur) and a higher multiplicity of the minimum in λ (Remark 5.3 shows that the hazard rate order may hold).

6 Two further examples

The first variant removes the ties that make the antiordering conditions of [14, 16] hold trivially for equal weights.

Proposition 6.1 (strict weights, distinct rates). *Let $p = (\frac{2}{5}, \frac{1}{3}, \frac{4}{15})$, $\lambda = (2, 6, 10)$ and $\gamma = (2, 7, 9)$, and let $T = \frac{3}{4}I + \frac{1}{4}\Pi_{23}$, so that $\gamma = \lambda T$ and $\tilde{p} := pT = (\frac{2}{5}, \frac{19}{60}, \frac{17}{60})$.*

- (a) *$\lambda \succ \gamma$; the weights p and $\tilde{p}$ are strictly decreasing, the rates λ, γ strictly increasing, so $(p_i - p_j)(\lambda_i - \lambda_j) < 0$, $(p_i - p_j)(\gamma_i - \gamma_j) < 0$ and $(\tilde{p}_i - \tilde{p}_j)(\gamma_i - \gamma_j) < 0$ for all $i \neq j$.*
- (b) *Common weights: $h_{p,\lambda}(\log 2) = \frac{898}{405}$ and $h_{p,\gamma}(\log 2) = \frac{214}{99}$, so $h_{p,\lambda}(\log 2) - h_{p,\gamma}(\log 2) = \frac{248}{4455} > 0$, while $h_{p,\lambda}(\log \frac{4}{3}) - h_{p,\gamma}(\log \frac{4}{3}) = -\frac{26862489}{459534895} < 0$.*
- (c) *Transformed weights: $h_{\tilde{p},\gamma}(\log 2) = \frac{6829}{3165}$, so $h_{p,\lambda}(\log 2) - h_{\tilde{p},\gamma}(\log 2) = \frac{1019}{17091} > 0$, while $h_{p,\lambda}(\log \frac{4}{3}) - h_{\tilde{p},\gamma}(\log \frac{4}{3}) = -\frac{399285261}{7327839955} < 0$.*
- (d) *In both cases $h_{p,\lambda}$ lies below the other hazard rate near $t = 0$ and above it for large t ; the exact number of sign changes on $(0, \infty)$ is one in each case, at $t \approx 0.380362$ for (b) and $t \approx 0.367980$ for (c).*
- (e) *$S_{p,\lambda}(t) - S_{p,\gamma}(t) = \frac{1}{15}q^6(1 - q)(5 - 4q^3) > 0$ for $t > 0$.*

Proof. (a) The decreasing rearrangements $(10, 6, 2)$ and $(9, 7, 2)$ have partial sums $10 \geq 9$, $16 \geq 16$, $18 = 18$. The T -transform maps $(6, 10)$ to $(7, 9)$ and $(\frac{1}{3}, \frac{4}{15})$ to $(\frac{1}{4} + \frac{1}{15}, \frac{1}{12} + \frac{1}{5}) = (\frac{19}{60}, \frac{17}{60})$.

(b), (c) With $q = \frac{1}{2}$, $S_{p,\lambda} = \frac{1}{10} + \frac{1}{192} + \frac{1}{3840} = \frac{405}{3840}$ and the numerator of $h_{p,\lambda}$ is $\frac{1}{5} + \frac{1}{32} + \frac{1}{384} = \frac{898}{3840}$; similarly $S_{p,\gamma} = \frac{198}{1920}$ with numerator $\frac{428}{1920}$, and $S_{\tilde{p},\gamma} = \frac{3165}{30720}$ with numerator $\frac{6829}{30720}$. The values at $q = \frac{3}{4}$ are rational numbers computed in the same way (Section 9).

(d) The p -means are $\mu_\lambda = \frac{82}{15} < \frac{83}{15} = \mu_\gamma$ and, for the transformed weights, $\frac{82}{15} = \frac{164}{30} < \frac{167}{30}$; both rate vectors have minimum 2 and second values $6 < 7$. The proof of Theorem 5.4 applies verbatim to two mixtures with different weight vectors, because Lemma 2.2 is applied to each mixture separately; it gives the sign pattern near 0 and near ∞ , hence at least one crossing in each case. For the exact count put $A = 6 + 5q^4 + 4q^8$, $C = 6 + 5q^5 + 4q^7$ and $\tilde{C} = 24 + 19q^5 + 17q^7$, which are positive on $(0, 1)$. Then

$$\begin{aligned} h_{p,\lambda} - h_{p,\gamma} &= \frac{q^4 P(q)}{AC}, & h_{p,\lambda} - h_{\tilde{p},\gamma} &= \frac{q^4 \tilde{P}(q)}{A\tilde{C}}, \\ P &= 16q^{11} + 60q^9 - 60q^7 - 25q^5 + 192q^4 - 168q^3 - 150q + 120, \\ \tilde{P} &= 68q^{11} + 228q^9 - 255q^7 - 95q^5 + 768q^4 - 714q^3 - 570q + 480. \end{aligned}$$

The substitution $q = 1/(1+x)$ maps $(0, 1)$ onto $x \in (0, \infty)$, and $Q(x) = (1+x)^{11}P(1/(1+x))$ has the coefficients, from x^{11} down to x^0 ,

$$120, 1170, 5100, 12882, 20448, 20555, 12114, 2625, -1628, -1497, -450, -15,$$

and $\tilde{Q}(x) = (1+x)^{11}\tilde{P}(1/(1+x))$ the coefficients

$$480, 4710, 20700, 52836, 85056, 87349, 53694, 13920, -5224, -5841, -1890, -90.$$

Each sequence has exactly one sign change, so by Descartes' rule of signs Q and $\tilde{Q}$ have exactly one positive root each, and it is simple. Hence P and $\tilde{P}$ have exactly one root in $(0, 1)$, where they change sign. The factorizations and the coefficients are checked exactly by the script (Section 9), which also confirms the counts by Sturm sequences.

$$(e) S_{p,\lambda} - S_{p,\gamma} = \frac{1}{3}(q^6 - q^7) + \frac{4}{15}(q^{10} - q^9) = \frac{1}{15}q^6(1-q)(5-4q^3). \quad \square$$

The second variant shows that the crossing does not require a common minimal rate, and that it can occur inside a bounded window. It is also an instance of a chain of two T -transforms with different structures, which is the setting of Theorem 3.9 of [14].

Proposition 6.2 (distinct minimal rates, two crossings). *Let $\lambda = (1, 3, 5)$ and $\gamma' = (\frac{41}{40}, \frac{159}{40}, 4)$ with equal weights. Then $\gamma' = \lambda T_1 T_2$ with $T_1 = \frac{1}{2}(I + \Pi_{23})$ and $T_2 = \frac{119}{120}I + \frac{1}{120}\Pi_{12}$, so $\lambda \succ (1, 4, 4) \succ \gamma'$, and:*

- (a) $h_\lambda(0) = h_{\gamma'}(0) = 3$ and $h'_\lambda(0) = -\frac{8}{3} < -\frac{4681}{2400} = h'_{\gamma'}(0)$, so $h_\lambda < h_{\gamma'}$ near 0;
- (b) $h_\lambda(t) \rightarrow 1 < \frac{41}{40} \leftarrow h_{\gamma'}(t)$, so $h_\lambda < h_{\gamma'}$ for large t ;
- (c) $h_\lambda(t_1) > h_{\gamma'}(t_1)$ at $t_1 = 40 \log \frac{500}{483} \approx 1.3838$, where the difference is a rational number computed exactly;
- (d) the difference changes sign exactly twice on $(0, \infty)$, at $t \approx 1.043788$ and $t \approx 1.885779$ (machine-certified), and $S_\lambda > S_{\gamma'}$ on $(0, \infty)$.

Proof. T_1 maps $(3, 5)$ to $(4, 4)$ and T_2 maps $(1, 4)$ to $(\frac{119+4}{120}, \frac{1+476}{120}) = (\frac{41}{40}, \frac{159}{40})$. (a) and (b) follow from Lemma 2.2: $\text{Var}(\gamma') = \frac{1}{3}(\frac{1681+25281}{1600} + 16) - 9 = \frac{4681}{2400}$. For (c) and (d), with $s = e^{-t/40}$ every term of (1) is a monomial in s with integer exponent, so the hazard difference is a rational function of s . Formed directly as $40(N_\lambda D_{\gamma'} - N_{\gamma'} D_\lambda)/(D_\lambda D_{\gamma'})$, with N/D the hazard rate of each mixture after division by its lowest power of s , the numerator has degree 279; after cancelling the factor $(1+s+s^2)^2$ common to numerator and denominator it has degree 275. Its value at $s = \frac{483}{500}$ is computed exactly and is positive, its values at $s = \frac{999}{1000}$ and $s = \frac{1}{2}$ are negative, and its distinct real roots in $(0, 1)$ are isolated exactly by the Vincent-Akritas-Strzeboński continued-fraction method as implemented in SymPy's `Poly.intervals` [1] (Section 9); there are exactly two, and the denominator has no root in $(0, 1)$. The signs $-, +, -$ at the three rational points (in increasing order of t) force a root of odd multiplicity in each of the two gaps between them, so each of the two distinct roots is a sign change. The survival difference is treated in the same way. $\square$

7 Relation to the published statements

7.1 Shojaee, Asadi and Finkelstein (2022)

The α -mixture of survival functions was introduced by Asadi, Ebrahimi and Soofi [2]. In [16], the generalized finite α -mixture of survival functions $\bar{F}_i$ is $\bar{F}(t, \bar{\alpha}) = [\sum_{i=1}^n p_i \bar{F}_i^{\alpha_i}(t)]^{1/\bar{\alpha}}$ with $\bar{\alpha} = \sum_i p_i \alpha_i$ (their equation (3)); $\alpha_i = 1$ for all i gives the arithmetic mixture. For a parametric family $\bar{F}(t | \lambda)$ with hazard rate $r(t | \lambda)$, $W_n(p, \lambda)$ denotes the mixture with parameters $\lambda_1, \dots, \lambda_n$, and (their page 1069)

$$\mathcal{U}_n = \{(x, y) : x_i, y_i \geq 0, (x_i - x_j)(y_i - y_j) \leq 0, i, j = 1, \dots, n\}.$$

Theorem 6.3 (page 1073) states: let $W_2(p, \lambda)$ and $W_2(p, \gamma)$ be two-component generalized finite α -mixtures with common mixing proportions p_1, p_2 , and suppose that for $(p, \lambda) \in \mathcal{U}_2$ and $(p, \gamma) \in \mathcal{U}_2$: (i) $r(t | \lambda)$ is increasing (decreasing) and concave (convex) in $\lambda > 0$ for all t ; (ii) $\bar{F}(t | \lambda)$ is decreasing (increasing) in $\lambda > 0$ for all t ; (iii) $\alpha_1 \leq \alpha_2$; (iv) $\lambda \succ \gamma$. Then, for $\alpha_i \geq 0$ such that $\alpha_1 p_1 \geq \alpha_2 p_2$ ($\alpha_i < 0$), $r_{W_2(p, \lambda)} \leq (\geq) r_{W_2(p, \gamma)}$. Remark 6.3 (page 1075) is quoted in Section 1; Example 6.4 illustrates the theorem with $\bar{F}(t | \lambda) = e^{-\lambda t}$.

The n -component statement whose validity Remark 6.3 leaves open is obtained by replacing 2 by n , (iii) by $\alpha_1 \leq \dots \leq \alpha_n$ and the weight condition by $\alpha_1 p_1 \geq \dots \geq \alpha_n p_n$, exactly as in Theorems 6.1 and 6.2 of [16], which treat the usual stochastic order for n components:

Statement 7.1 (the extension asked for in Remark 6.3). Let $(p, \lambda), (p, \gamma) \in \mathcal{U}_n$, let $\bar{F}(t | \lambda)$ satisfy (i) and (ii), let $\alpha_1 \leq \dots \leq \alpha_n$ with $\alpha_i \geq 0$ and $\alpha_1 p_1 \geq \dots \geq \alpha_n p_n$, and let $\lambda \succ \gamma$. Then $r_{W_n(p, \lambda)} \leq r_{W_n(p, \gamma)}$ on $[0, \infty)$.

For $\alpha_i = 1$ and $\bar{F}(t | \lambda) = e^{-\lambda t}$, conditions (i) and (ii) hold ($r(t | \lambda) = \lambda$ is increasing and concave, $e^{-\lambda t}$ is decreasing in λ), $W_n(p, \lambda)$ is the exponential mixture with weights p and rates λ , and the hypotheses reduce to $(p, \lambda), (p, \gamma) \in \mathcal{U}_n$, $p_1 \geq \dots \geq p_n$ and $\lambda \succ \gamma$; the conclusion is $h_{p, \lambda} \leq h_{p, \gamma}$. Statement 7.1 is therefore false for every $n \geq 3$: Theorem 3.1 and Theorem 4.1 give counterexamples with uniform p (for which all antiordering conditions hold with equality), and Proposition 6.1(b) gives a counterexample with strictly decreasing weights and strictly increasing rates, so that every inequality in the hypotheses is strict. The answer to Remark 6.3 is negative, already for ordinary mixtures of exponential distributions, and no strengthening of the antiordering condition repairs it. Theorem 6.3 itself ($n = 2$) is not affected; Theorem 5.1(c) with $w = 0$ and Remark 4.2 are consistent with it.

Minor errata and a remark in [16]. The exact-arithmetic audit described in Section 9 checked the remaining finitely testable claims of both manuscripts; three items in [16] are worth recording here, none of which contradicts a main claim.

(i) *Theorem 4.1(b)*. In the multiplicative model $r_i(t) = \lambda_i r(t)$ with $\lambda_1 \leq \lambda_2$, $r(t) \rightarrow \infty$, $\alpha_i > 0$, that statement asserts $r(t, \bar{\alpha}) - (\alpha_1 / \bar{\alpha}) r_1(t) \rightarrow 0$ as $t \rightarrow \infty$ if and only if $r(t) \exp(-(\alpha_2 \lambda_2 - \alpha_1 \lambda_1) \int_0^t r(u) du) \rightarrow 0$, under the hypothesis $\alpha_1 \leq c \alpha_2$ where $c = \lambda_2 / \lambda_1$. The “only if” direction fails on the boundary $\alpha_1 \lambda_1 = \alpha_2 \lambda_2$, which the hypothesis permits: there $\sum_i p_i \bar{F}_i^{\alpha_i}(t) = e^{-\alpha_1 \lambda_1 \int_0^t r}$ identically, hence $r(t, \bar{\alpha}) \equiv (\alpha_1 / \bar{\alpha}) r_1(t)$ and the difference is identically 0, while the displayed condition reduces to $r(t) \rightarrow 0$, incompatible with the standing assumption $r(t) \rightarrow \infty$. (The simplest instance is $\lambda_1 = \lambda_2$ and $\alpha_1 = \alpha_2$.) Replacing the hypothesis by the strict inequality $\alpha_1 < c \alpha_2$, i.e. $\alpha_1 \lambda_1 < \alpha_2 \lambda_2$, repairs the biconditional.

(ii) *Example 3.6*. For $\bar{F}_1(t) = e^{-0.8t}$, $\bar{F}_2(t) = e^{-0.4t}$, $(p_1, p_2) = (0.7, 0.3)$ and $(\alpha_1, \alpha_2) = (2, 5)$, the hazard rate printed there,

$$r(t, \bar{\alpha}) = \frac{1}{2.9} \cdot \frac{1.12 e^{-1.6t} + 0.6 e^{-2t}}{e^{-1.6t} + e^{-2t}},$$

omits the weights p_i in the denominator, which should read $0.7e^{-1.6t} + 0.3e^{-2t}$: the printed formula gives $r(0, \bar{\alpha}) = 43/145 \approx 0.297$ while the correct value is $86/145 \approx 0.593$. The conclusion drawn in the example ($r(t, \bar{\alpha}) < r_1(t) = 0.8$) is unaffected.

(iii) *Theorem 5.2, condition (i)*, for $\alpha_i < 0$. The printed hypothesis that $(\alpha_i / \bar{\alpha}) r_{F_i}(t)$ be increasing in i is, since $\bar{\alpha} < 0$, equivalent to $\alpha_i r_{F_i}(t)$ decreasing in i ; a reader who replaces it by “ $\alpha_i r_{F_i}$ increasing” will construct apparent counterexamples to the claim. Under the correct reading the claim held on all 5741 admissible instances tested in the audit (a bounded exact-rational search, not a proof). The statement would benefit from spelling out the $\alpha_i < 0$ case.

7.2 Sahoo, Kayal and Finkelstein (2026)

Statement numbers, page numbers and quotations below refer to the peer-reviewed accepted author manuscript (Strathprints record 96227); Section 8 describes this source and its relation to the version of record. The finite α -mixture with resilience-scaled components is (their equation (1.3), page 3)

$$\bar{G}_{U_n(p,\gamma,\theta)}(t) = \left[\sum_{i=1}^n p_i \bar{G}^{\alpha\gamma_i}(t/\theta_i) \right]^{1/\alpha} \quad (\alpha \neq 0),$$

where $\bar{G}$ is a baseline survival function with hazard rate h , $\gamma_i > 0$ are resilience parameters and $\theta_i > 0$ scale parameters; in Theorems 3.7 to 3.12 the scale is a common scalar θ . With $\bar{G}(t) = e^{-t}$, $\theta = 1$ and $\alpha = 1$ this is exactly $S_{p,\gamma}$ in (1), and the hazard rate displayed in the proof of Theorem 3.8, $h_{U_n(p,\gamma,\theta)}(t) = \theta^{-1}g(t/\theta) \sum_i p_i \gamma_i \bar{G}^{\alpha\gamma_i-1}(t/\theta) / \sum_i p_i \bar{G}^{\alpha\gamma_i}(t/\theta)$, reduces to $h_{p,\gamma}$. The classes (page 6) are

$$\mathcal{V}_n = \left\{ \begin{bmatrix} y_1 & \cdots & y_n \\ z_1 & \cdots & z_n \end{bmatrix} : y_i, z_j > 0, (y_i - y_j)(z_i - z_j) \leq 0 \right\},$$

and $\mathcal{W}_n$ with ≥ 0 ; $Y \leq_{\text{hr}} Z$ means $h_Y \geq h_Z$ (Definition 2.1(ii), page 4).

Statement 7.2 (Theorem 3.8 of [14], accepted manuscript, page 20). “Let $U_n(p, \gamma, \theta)$ and $V_n(q, \delta, \theta)$ be the finite α -mixture random variables. Suppose $h(t)$ is positive and M_T is a T -transform matrix. Then, for $\begin{bmatrix} p \\ \gamma \end{bmatrix} \in \mathcal{V}_n$ (or $\in \mathcal{W}_n$) and $\alpha \geq 0$ (or ≤ 0),

$$\begin{bmatrix} q \\ \delta \end{bmatrix} = \begin{bmatrix} p \\ \gamma \end{bmatrix} M_T \implies U_n(p, \gamma, \theta) \geq_{\text{hr}} (\leq_{\text{hr}}) V_n(q, \delta, \theta).”$$

Take $\bar{G}(t) = e^{-t}$ (so $h \equiv 1 > 0$), $\theta = 1$ and $\alpha = 1$. The following instances satisfy every hypothesis of Statement 7.2 for the class $\mathcal{V}_n$ and contradict its conclusion $h_{p,\gamma} \leq h_{q,\delta}$:

- (1) $n = 3$, p uniform, $\gamma = (1, 3, 5)$, $M_T = \frac{1}{2}(I + \Pi_{23})$: then $q = p$, $\delta = (1, 4, 4)$, and $h_{p,\gamma}(\log 4) - h_{q,\delta}(\log 4) = \frac{41}{1001} > 0$ (Theorem 3.1).
- (2) Any $n \geq 3$, p uniform, $\gamma = \lambda^{(n-2)}$, $M_T = \frac{1}{2}(I + \Pi_{n-1,n})$: the difference at $t = \log 4$ is $3(128m - 5)/((256m + 17)(32m + 1)) > 0$ with $m = n - 2$ (Theorem 4.1).
- (3) $n = 3$, $p = (\frac{2}{5}, \frac{1}{3}, \frac{4}{15})$, $\gamma = (2, 6, 10)$, $M_T = \frac{3}{4}I + \frac{1}{4}\Pi_{23}$: then $q = (\frac{2}{5}, \frac{19}{60}, \frac{17}{60})$, $\delta = (2, 7, 9)$, $\begin{bmatrix} p \\ \gamma \end{bmatrix} \in \mathcal{V}_3$ with all products $(p_i - p_j)(\gamma_i - \gamma_j)$, $i \neq j$, strictly negative, and $h_{p,\gamma}(\log 2) - h_{q,\delta}(\log 2) = \frac{1019}{17091} > 0$ (Proposition 6.1(c)).

Corollary 3.2 there (a finite product $M_{T_1} \cdots M_{T_k}$ of T -transforms with the same structure) contains Statement 7.2 as the case $k = 1$ and is contradicted by the same instances. Theorem 3.9 there (a product of $k \geq 2$ T -transforms with different structures, with the intermediate matrices required to lie in $\mathcal{V}_n$) is contradicted by instance (4): p uniform, $\gamma = (1, 3, 5)$, $M_{T_1} = \frac{1}{2}(I + \Pi_{23})$, $M_{T_2} = \frac{119}{120}I + \frac{1}{120}\Pi_{12}$, for which the intermediate matrix has rows p and $(1, 4, 4)$ and lies in $\mathcal{V}_3$, $\delta = (\frac{41}{40}, \frac{159}{40}, 4)$, and $h_{p,\gamma} > h_{q,\delta}$ at $t_1 = 40 \log \frac{500}{483}$ (Proposition 6.2(c)).

Where the published argument stops. The proof of Theorem 3.8 sets $\Psi_n(\begin{bmatrix} p \\ \gamma \end{bmatrix}) = h_{U_n(p,\gamma,\theta)}(t)$, observes that by Theorem 3.7 the two-column function Ψ_2 satisfies the conditions of Lemma 2.4 (a criterion for 2×2 matrices), and concludes “using Lemma 2.5”. Lemmas 2.4 and 2.5 are both attributed there to Barmalzan, Kosari and Balakrishnan [3], and Lemma 2.5 is stated for functions of the additive form $\Psi_n(A) = \sum_{i=1}^n \Psi(a_{1i}, a_{2i})$; the lemma itself is not in question. The hazard rate of a mixture is a ratio of two such sums and is not of this form, so the lemma does not apply to Ψ_n ; since the conclusion is false, no other argument can close the gap. Theorem 3.7 ($n = 2$) is not contradicted by anything in this note. The proof of Theorem 3.9 is said to follow from Theorem 3.4 of [8], which we could not retrieve (Section 8); we make no claim about that result beyond the fact that Theorem 3.9 as stated fails for instance (4).

The $\mathcal{W}_n$, $\alpha \leq 0$ halves. The instances (1) to (4) all lie in $\mathcal{V}_n$ with $\alpha > 0$. The complementary halves of the same statements (the class $\mathcal{W}_n$ with $\alpha \leq 0$) fail as well, on the following two instances, both certified in exact rational arithmetic by the audit scripts of Section 9.

(5) $n = 3$; $p = (\frac{17}{44}, \frac{5}{11}, \frac{7}{44})$ and $\gamma = (7, 9, 2)$ are comonotone, so $[\frac{p}{\gamma}] \in \mathcal{W}_3$; $M_T = \frac{17}{20}I + \frac{3}{20}\Pi_{13}$ gives $q = (\frac{31}{88}, \frac{5}{11}, \frac{17}{88})$ and $\delta = (\frac{25}{4}, 9, \frac{11}{4})$; $\bar{G}(t) = e^{-t}$, $\theta = 1$, $\alpha = -2$. By (12) below, $h_{U_3} - h_{V_3} = \tilde{h}_{p,\gamma}(y) - \tilde{h}_{q,\delta}(y)$ with $y = e^{2t} > 1$. Writing $w = e^{-t/2}$, the difference is the rational function

$$h_{U_3}(t) - h_{V_3}(t) = -\frac{w^8 B(w)}{D(w)}, \quad (10)$$

$$\begin{aligned} B(w) &= 5440 - 6820w^3 - 1581w^{11} - 8500w^{17} + 7840w^{20} - 4913w^{25} + 3689w^{31} + 357w^{45}, \\ D(w) &= 3200 + 2720w^8 + 2480w^{11} + 2108w^{19} + 1360w^{25} + 1120w^{28} + 1156w^{33} + 868w^{39} + 476w^{53} > 0. \end{aligned}$$

Now $B(0) = 5440 > 0$ and $B(1) = -4488 < 0$, and B has exactly one root w_* in $(0, 1)$ (machine-certified, in $(\frac{7}{8}, \frac{9}{10})$ since $B(\frac{7}{8}) > 0 > B(\frac{9}{10})$). Hence $h_{U_3} - h_{V_3}$ is positive for $t < -2\log w_*$ and negative beyond it: the claimed $U_3 \leq_{\text{hr}} V_3$ fails, the hazard rates cross exactly once, and the $\mathcal{W}_n$, $\alpha \leq 0$ half of Statement 7.2 is false. Exact witnesses: the difference equals $-161379121858494821/28929682881644134876 < 0$ at $e^{-t} = \frac{1}{4}$ and is positive (≈ 0.16358 , an exact algebraic value) at $e^{-t} = \frac{7}{8}$. The same single-transform instance contradicts the $\mathcal{W}_n$ half of Corollary 3.2 (the case $k = 1$).

For Theorem 3.9 a single T -transform does not suffice: that statement requires a product of $k \geq 2$ transforms with different structures and intermediate matrices in the class. A certified two-step chain is

(6) $p = (\frac{7}{26}, \frac{4}{13}, \frac{11}{26})$, $\gamma = (3, 8, 9)$, so $[\frac{p}{\gamma}] \in \mathcal{W}_3$; $M_{T_1} = \frac{1}{20}I + \frac{19}{20}\Pi_{23}$ gives $[\frac{q_1}{\gamma_1}] = [\frac{7/26}{3}, \frac{217/520}{179/20}, \frac{163/520}{161/20}] \in \mathcal{W}_3$; then $M_{T_2} = \frac{1}{4}I + \frac{3}{4}\Pi_{13}$ gives $q = (\frac{629}{2080}, \frac{217}{520}, \frac{583}{2080})$ and $\delta = (\frac{543}{80}, \frac{179}{20}, \frac{341}{80})$; $\bar{G}(t) = e^{-t}$, $\theta = 1$, $\alpha = -1$. In $w = e^{-t/80}$ the difference $h_{U_3} - h_{V_3}$ is a rational function whose numerator has degree 855; its exact rational values are $\frac{7881}{83200} > 0$ at $w = 1$ (i.e. $t = 0$), negative (≈ -0.02834) at $w = \frac{49}{50}$, and $\frac{1}{20} > 0$ at $w = \frac{1}{2}$. The claimed order therefore fails on an interior band around $t \approx 1.6$, and the signs $+, -, +$ force at least two crossings: the $\mathcal{W}_n$, $\alpha \leq 0$ half of Theorem 3.9 is false as well.

Theorem 3.4 fails for $\alpha > 0$. Theorem 3.4 of [14] (page 12) is a usual-stochastic-order statement in the scale parameters: if $t^2 g(t)$ is increasing in $t > 0$, then for $\alpha < \gamma_1$ with $p \in \varepsilon_n^+$ (or $\mathcal{D}_n$), $\theta, \xi \in \mathcal{D}_n^+$ (or ε_n^+) and $\theta \prec^w \xi$ (the increasing partial sums of θ dominate those of ξ), one has $U_n(p, \gamma, \theta) \geq_{\text{st}} V_n(p, \gamma, \xi)$. Take $n = 2$, scalar $\gamma = 3$, $\alpha = \frac{2}{3} < \gamma$, $p = (\frac{1}{3}, \frac{2}{3}) \in \varepsilon_2^+$, and $\theta = (3, 1)$, $\xi = (\frac{7}{2}, \frac{1}{2}) \in \mathcal{D}_2^+$; the weak supermajorization holds since $1 \geq \frac{1}{2}$ and $4 = 4$. For the Lomax baseline $\bar{G}(t) = 1/(1+t)$, for which $t^2 g(t) = t^2/(1+t)^2$ is strictly increasing in $t > 0$, the sign of $\bar{G}_{U_2} - \bar{G}_{V_2}$ equals the sign of the inner difference

$$d(t) = \sum_i p_i \left[\left(\frac{\theta_i}{\theta_i + t} \right)^2 - \left(\frac{\xi_i}{\xi_i + t} \right)^2 \right] = \frac{t(1722 + 4565t + 3664t^2 + 1053t^3 + 4t^4 - 28t^5)}{3(t+1)^2(t+3)^2(2t+1)^2(2t+7)^2}, \quad (11)$$

since $\alpha\gamma = 2$ and $\alpha > 0$. The quintic factor has a single sign change in its coefficient sequence, so exactly one positive root by Descartes' rule, lying in $(7, 8)$ because the factor equals 113400 at $t = 7$ and -89246 at $t = 8$. Hence $d(t) > 0$ on $(0, t_*)$ and $d(t) < 0$ on (t_*, ∞) ; exact witnesses are $d(1) = 305/3888 > 0$ and $d(20) = -1579547560/2598836427243 < 0$. The claimed $U_2 \geq_{\text{st}} V_2$ therefore fails for large t , and since the difference is positive near the origin the reverse $V_2 \geq_{\text{st}} U_2$ fails too: the two mixtures are stochastically incomparable. (A three-component instance is certified by the same script: $p = (\frac{25}{86}, \frac{27}{86}, \frac{17}{43})$, $\gamma = 6$, $\alpha = \frac{5}{6}$, $\theta = (8, 8, 5)$, $\xi = (9, 8, 4)$.) The failure is located in the Schur-concavity step of the published proof (its equation (3.7)), where factors whose signs compete, since $\bar{G}^{\alpha\gamma-1}(t/\theta)$ and $(t/\theta)^2 g(t/\theta)/\theta^2$ move in opposite directions for $\theta_i > \theta_j$ depending on the sign of $\alpha\gamma - 1$, are combined as if the displayed sign followed from the hypotheses. The audit's sampled violations occur on both sides of $\alpha\gamma = 1$, so an $\alpha\gamma \leq 1$ restriction does not repair the statement; the $\alpha < 0$ half held on all 589 admissible sampled instances (bounded search, not a proof). The proof of Theorem 3.6 there invokes Theorem 3.4 only in that $\alpha \leq 0$ regime, so the counterexample does not propagate to it.

Two invalid counterexamples printed in the manuscript. Counterexamples 3.7 and 3.8 of [14] (page 16) are offered to show the need for the hypotheses of Theorem 3.6; their numerical content is wrong in both cases.

Counterexample 3.7 prints $\bar{G}_{U_3}(20) - \bar{G}_{V_3}(20) = -0.08336543$ for $p = (0.05, 0.3, 0.65)$, $q = (0.7, 0.26, 0.4)$, $\gamma = 2$, $\theta = (5, 3.5, 2)$, $\xi = (5, 3, 1.5)$, $\alpha = -0.3$ and the Pareto baseline $\bar{G}(t) = t^{-1/2}$; recomputation at 60-digit precision gives $+0.06744747$: the difference is positive at $t = 20$, in the direction claimed by Theorem 3.6, and the printed number is simply erroneous. Counterexample 3.8 reports $h_{U_3} - h_{V_3}$ taking the values $-1.110223 \times 10^{-16}$ and $+2.220446 \times 10^{-16}$ at $t = 6$ and $t = 7$; but for the Pareto baseline with scalar γ the hazard rate displayed there simplifies to $h_{U_3}(t) = \gamma/(2t) = h_{V_3}(t)$ identically, so the two hazard rates are equal and the reported values are floating-point roundoff.

7.3 Scope, and the reversed hazard rate statements

For $y > 0$ put $\tilde{h}_{p,\gamma}(y) = \sum_i p_i \gamma_i y^{\gamma_i} / \sum_i p_i y^{\gamma_i}$, so that $h_{p,\gamma}(t) = \tilde{h}_{p,\gamma}(e^{-t})$. Differentiating (1.3) with a common scale θ and $\alpha \neq 0$ gives

$$h_{U_n(p,\gamma,\theta)}(t) = \theta^{-1} h(t/\theta) \tilde{h}_{p,\gamma}(\bar{G}(t/\theta)^\alpha), \quad (12)$$

which is the expression displayed in the proof of Theorem 3.8.

Scope of the failure of Theorem 3.8. Let $\bar{G}$ be continuous with $\bar{G}(0) = 1$, $\bar{G}(t) \rightarrow 0$ as $t \rightarrow \infty$ and positive hazard rate h . For $\alpha > 0$ the value $y = \bar{G}(t/\theta)^\alpha$ runs through all of $(0, 1)$ as t runs through $(0, \infty)$, and by (12) the sign of $h_{U_n(p,\gamma,\theta)}(t) - h_{V_n(q,\delta,\theta)}(t)$ is the sign of $\tilde{h}_{p,\gamma}(y) - \tilde{h}_{q,\delta}(y)$, that is, of the exponential-mixture difference $h_{p,\gamma} - h_{q,\delta}$ at time $-\log y$. Hence instances (1) to (4) contradict Theorem 3.8, Corollary 3.2 and Theorem 3.9 for every such baseline, every $\theta > 0$ and every $\alpha > 0$, not only for $\bar{G}(t) = e^{-t}$ and $\theta = \alpha = 1$. The half of these statements with $\mathcal{W}_n$ and $\alpha \leq 0$ corresponds to the range $y = \bar{G}^\alpha > 1$, which the instances (1) to (4) do not reach: for instance (1), whose uniform weights also lie in $\mathcal{W}_3$, writing $y = 1/q$,

$$\tilde{h}_{(1,3,5)}(1/q) - \tilde{h}_{(1,4,4)}(1/q) = \frac{2(q-1)(q^4 - 2q^3 - q - 1)}{(q^3 + 2)(q^2 - q + 1)(q^2 + q + 1)},$$

which is positive on $(0, 1)$ because $q^4 - 2q^3 - q - 1 < q^4 - 1 < 0$ there; this is the direction that the $\alpha \leq 0$ half predicts, and on this instance it holds. It does not hold in general: the comonotone instance (5) of Section 7.2 has a certified sign change in the same range $y > 1$, so the $\mathcal{W}_n$, $\alpha \leq 0$ half of Theorem 3.8 fails as well, and both halves of the statement are false.

Reversed hazard rates. Theorem 3.11 of [14] (page 23) reads: “Let $U_n(p, \gamma, \theta)$ and $V_n(q, \delta, \theta)$ be the finite α -mixture random variables. Suppose $r(t)$ is positive and M_T is a T -transform matrix. Then, for $\begin{bmatrix} p \\ \gamma \end{bmatrix} \in \mathcal{V}_n$ (or $\in \mathcal{W}_n$) and $\alpha \geq 0$ (or ≤ 0). $\begin{bmatrix} q \\ \delta \end{bmatrix} = \begin{bmatrix} p \\ \gamma \end{bmatrix} M_T \implies U_n(p, \gamma, \theta) \leq_{\text{rh}} (\geq_{\text{rh}}) V_n(q, \delta, \theta)$.” Here $r = g/G$ is the reversed hazard rate of the baseline distribution function $G = 1 - \bar{G}$. Corollary 3.3 (page 24) is the same statement for a product $M_{T_1} \cdots M_{T_k}$ of T -transforms with the same structure, and Theorem 3.12 (page 24) is the version for $k \geq 2$ T -transforms with different structures and intermediate matrices in $\mathcal{V}_n$ (or $\mathcal{W}_n$), whose proof is again deferred to Theorem 3.4 of [8]. The proof of Theorem 3.11 displays the reversed hazard rate of U_n as

$$\frac{\theta^{-1} g(t/\theta) \sum_i p_i \gamma_i G^{\alpha \gamma_i - 1}(t/\theta)}{\sum_i p_i G^{\alpha \gamma_i}(t/\theta)} = \theta^{-1} r(t/\theta) \tilde{h}_{p,\gamma}(G(t/\theta)^\alpha), \quad (13)$$

and refers to the argument of Theorem 3.8 “with the help of Lemma 2.5”. The model (1.3) admits two readings of the statement, and the instances contradict it under both.

(a) *With the displayed formula* (13). The conclusion $r_{U_n} \leq r_{V_n}$ for $\mathcal{V}_n$ and $\alpha \geq 0$ becomes $\tilde{h}_{p,\gamma}(y) \leq \tilde{h}_{q,\delta}(y)$ at $y = G(t/\theta)^\alpha$, which for $\alpha > 0$ runs through $(0, 1)$; this is literally the comparison $h_{p,\gamma} \leq h_{q,\delta}$ asserted by Theorem 3.8. So instances (1) to (3) contradict Theorem 3.11 and Corollary 3.3, and instance (4) contradicts Theorem 3.12, for every baseline with G continuous from 0 to 1 and $r > 0$, every $\theta > 0$ and every $\alpha > 0$. The same reduction applies on $y = G(t/\theta)^\alpha > 1$ for $\alpha \leq 0$: under this reading instance (5) contradicts the $\mathcal{W}_n$ halves of Theorem 3.11 and Corollary 3.3, and instance (6) contradicts the $\mathcal{W}_n$ half of Theorem 3.12. For example, with $G(t) = 1 - e^{-t}$, $\theta = \alpha = 1$ and instance (1), at $t = \log \frac{4}{3}$ one has $G = \frac{1}{4}$, $r = 3$, and (13) gives $r_{U_3} - r_{V_3} = 3(\frac{103}{91} - \frac{12}{11}) = \frac{123}{1001} > 0$. We note that (13) is the reversed hazard rate of the distribution function $[\sum_i p_i G^{\alpha \gamma_i}(t/\theta)]^{1/\alpha}$, an α -mixture of the distribution functions $G^{\gamma_i}(t/\theta)$, and not the reversed hazard rate of the survival model (1.3): in the example just given at $t = \log 2$, (13) equals $\frac{11}{7}$ for $\gamma = (1, 3, 5)$, whereas f/F for (1.3) equals $\frac{11}{25}$. We have checked this only against the accepted manuscript.

(b) *With the reversed hazard rate of (1.3).* For $\bar{G}(t) = e^{-t}$ and $\theta = \alpha = 1$, U_n is the exponential mixture and $r_{p,v}(t) = \sum_i p_i v_i q^{v_i} / (1 - \sum_i p_i q^{v_i})$ with $q = e^{-t}$. For instance (1),

$$r_{(1,3,5)} - r_{(1,4,4)} = \frac{q^3(9 - 8q - 4q^2 - 4q^3 - 2q^4)}{(3 + 2q + 2q^2 + q^3 + q^4)(3 + 2q + 2q^2 + 2q^3)},$$

which equals $\frac{11}{25} - \frac{8}{19} = \frac{9}{475} > 0$ at $t = \log 2$, contradicting $r_{U_3} \leq r_{V_3}$; so Theorem 3.11 and Corollary 3.3 fail under this reading too. Under this reading, however, the failure is not specific to $n \geq 3$. Theorems 3.7 and 3.10 of [14] would give $U_2 \geq_{\text{hr}} V_2$ and $U_2 \leq_{\text{rh}} V_2$, hence $U_2 \geq_{\text{st}} V_2$ and $U_2 \leq_{\text{st}} V_2$, so U_2 and V_2 would have the same distribution; this is false for the pair (3, 5), (4, 4) with equal weights, for which $S_{(3,5)} - S_{(4,4)} = \frac{1}{2}q^3(1 - q)^2 > 0$ and, directly, $r_{(3,5)}(\log 2) = \frac{17}{59} > \frac{4}{15} = r_{(4,4)}(\log 2)$. Since Theorem 3.7 holds for this pair (Remark 4.2), Theorem 3.10 cannot hold under reading (b). The same argument shows that, under reading (b), Theorems 3.8 and 3.11 cannot both hold for any admissible pair for which U_n and V_n are not equal in distribution. The failure under this reading is generic rather than exceptional: in the audit's exact-arithmetic bounded search over admissible two-component instances of the $\mathcal{V}_2$, $\alpha > 0$ branch (exponential baseline, $\theta = 1$, T -transformed pairs), 2279 of 2500 violate the literal reversed-hazard conclusion of Theorem 3.10.

We therefore regard (a) as the substantive reading: under it, the two-component Theorem 3.10 reduces to a two-component hazard rate comparison, which the audit confirmed on all admissible sampled instances (2×3000 tested instances across the two branches in the shipped default run; a bounded-search confirmation, not a proof), while Theorem 3.11, Corollary 3.3 and Theorem 3.12 fail for $n \geq 3$ at exactly the instances that refute Theorem 3.8, Corollary 3.2 and Theorem 3.9: instances (1) to (4) for the $\mathcal{V}_n$, $\alpha \geq 0$ halves and instances (5) and (6) for the $\mathcal{W}_n$, $\alpha \leq 0$ halves.

7.4 What survives

Proposition 7.3. *Let p be uniform, $\lambda \succ \gamma$, and γ not a rearrangement of λ .*

- (a) $S_\lambda(t) > S_\gamma(t)$ for every $t > 0$; the mixture with rates λ is strictly larger in the usual stochastic order.
- (b) There is $\varepsilon > 0$ with $h_\lambda < h_\gamma$ on $(0, \varepsilon]$: the conclusion of the two-component statements holds on a neighbourhood of the origin in every dimension.
- (c) $h_\lambda(t) \rightarrow \lambda_{(1)}$ and $h_\gamma(t) \rightarrow \gamma_{(1)}$ with $\lambda_{(1)} \leq \gamma_{(1)}$; both hazard rates are strictly decreasing.
- (d) For $n = 2$, $h_\lambda \leq h_\gamma$ on $[0, \infty)$.

Proof. (a) Apply Lemma 2.3(a) to the strictly convex function $x \mapsto e^{-xt}$, $t > 0$. (b) Majorization gives equal means and Lemma 2.3(b) gives $\text{Var}(\lambda) > \text{Var}(\gamma)$; apply Lemma 2.2(a). (c) Lemma 2.2(b),(c) and $\lambda_{(1)} \leq \gamma_{(1)}$ from Definition 2.1. (d) For $n = 2$, $\lambda = (b - d, b + d)$ and $\gamma = (b - d', b + d')$ with $0 \leq d' < d$, and $h_\lambda(t) = b - d \tanh(dt) \leq b - d' \tanh(d't) = h_\gamma(t)$ because $x \mapsto x \tanh(xt)$ is increasing on $[0, \infty)$. $\square$

Part (a) is the exponential case of Theorem 6.1 of [16] (usual stochastic order, n components), which our examples do not contradict. Part (b) explains why numerical checks on a short time interval cannot detect the failure: in Theorem 4.1 the crossing time is above $\log 2$ and the sign change happens after the hazard rates have already dropped to within a small distance of their common limit. For completeness we record the audit's consistent findings (all bounded exact samples, none a proof): the two-component statements of [14] (Theorem 3.7 in all four quarters and Theorem 3.10 under the displayed-formula reading) held on every admissible instance tested, as did the $\alpha < 0$ half of its Theorem 3.4, its stochastic-order statements Theorems 3.1 to 3.3 and 3.5 to 3.6, and the stochastic-order and hazard statements of [16] tested in that paper's parameter families, including its Theorems 6.1 to 6.3.

8 Sources, priority and literature search

Sources consulted. All statements attributed to [16] were read in the publisher's version of record (Cambridge Core PDF, 25 pages, downloaded 25 September 2026; hash below): Definition 6.1 on page 1068, U_n and Theorem 6.1 on page 1069, Theorem 6.3 on page 1073, Remark 6.3 and Example 6.4 on page 1075, and Theorem 4.1, Example 3.6 and Theorem 5.2 in the same source. All statements attributed to [14] were read in the

peer-reviewed accepted author manuscript deposited in the University of Strathclyde repository (Strathprints record 96227, 26 pages, whose cover sheet identifies it as the accepted manuscript of the article in volume 42, number 2, article e70089; deposited 12 May 2026; hash below): equation (1.3) on page 3, Definition 2.1 on page 4, $\mathcal{V}_n$, $\mathcal{W}_n$, Lemmas 2.4 and 2.5 on page 6, Theorem 3.4 on page 12, Theorem 3.5 on page 14, Theorem 3.6 and Counterexamples 3.7 and 3.8 on page 16, Theorem 3.7 on page 17, Theorem 3.8, Corollary 3.2, Theorem 3.9 and Theorem 3.10 on page 20, Theorem 3.11 on page 23, and its proof, Corollary 3.3 and Theorem 3.12 on page 24. The publisher’s version of record could not be retrieved (access restricted), and the Crossref record of the article listed no correction or update on 25 September 2026; statement numbers or wording in the version of record may differ from the accepted manuscript, and the reader should check them there.

Sources not retrieved. The full texts of [8] and [11] were not accessible to us; their bibliographic records were verified through Crossref, and they are cited only because Remark 6.3 of [16] and the proofs of Theorems 3.9 and 3.12 of [14] refer to them. A further attempt on 25 September 2026 to locate an open copy of [8] found only the publisher’s page. Whether they contain n -component hazard-rate statements affected by our examples is unknown to us. The paper [2] is cited only as the origin of the α -mixture model, as both [16] and [14] attribute it; its bibliographic record was verified through Crossref and its text was not examined. The article [13] was seen only through its Crossref record and abstract (the publisher’s page was inaccessible) and was not examined. Of [4] only the abstract was seen; it concerns preservation of stochastic orders under exponential mixtures ordered through the mixing distribution, a different question, and is cited for context only. The arXiv preprint [7] was retrieved and read (hash below); its hazard-rate results (Theorems 5 and 6, Corollaries 5 and 6) vary the tilt parameters of a modified proportional hazard rate model at a common proportional-hazards parameter, so the exponential-rate mixtures of this note are not instances of them, and nothing here bears on that paper.

Neighbouring results. Stochastic comparisons of finite mixtures in neighbouring models are the subject of, for example, Barmalzan, Kosari and Balakrishnan [3] (location-scale components; the source of Lemmas 2.4 and 2.5 of [14]), Bhakta, Kayal and Balakrishnan [6] (multiple-outlier finite mixtures), Guo and Yan [7] and Sahoo, Kayal and Balakrishnan [13]. Apart from the reading of [7] reported above, we have not examined these papers beyond their bibliographic records (verified through Crossref) and, for [13], its abstract; in particular we have not checked whether any of their n -component hazard rate or reversed hazard rate results applies an additive lemma of the type of Lemma 2.5 to a functional that is not additive over columns. They are outside the scope of this note, and nothing here is claimed about them.

Retrieved files (SHA-256).

6750bb3ec436b230b2caae9b38a3f801e847609988af0084176ec0c475e3256b
version of record of [16], Cambridge Core PDF, 25 pages.

67efbc3770e16abdb2c4c2c6158323bf8e4966e9d8dd4dae6d0ac8efd6d1fe7c
accepted author manuscript of [14], Strathprints record 96227, 26 pages.

26cbe4d4c562c0af4be0b852b694ec20e98c86dde01d40691d2295b2780bec0d
arXiv:2407.15638v2 [7], 32 pages.

Literature search. On 25 September 2026 we ran eight web searches¹ followed by targeted retrievals of the publisher, repository and arXiv pages of the items found (including arXiv:2511.00791, whose abstract claims no hazard-rate result under majorization). None of the results contained a counterexample to the n -component hazard-rate statement, a correction of [14], or an arXiv version of [16]. The search was bounded and we make no claim that it was exhaustive.

¹The queries were: “Shojaee Asadi Finkelstein generalized finite α -mixtures arXiv stochastic properties”; “hazard rate order finite mixture majorization counterexample exponential mixtures”; “mixture of exponential distributions hazard rate crossing majorization Schur-convex mixture hazard rate rates vector”; “Bartoszewicz Mixtures of exponential distributions and stochastic orders Statistics Probability Letters”; “Sahoo Kayal Finkelstein resilience-scaled α -mixture stochastic ordering arXiv OR corrigendum OR correction”; “finite mixtures hazard rate order majorization $n > 2$ OR more than two open problem counterexample exponential components”; “Shojaee generalized finite α -mixtures Remark 6.3 OR open problem hazard rate order n -component mixtures majorization”; “chain majorization T-transform hazard rate order finite mixture counterexample does not hold”.

9 Reproducibility

Every numerical statement in this note was re-derived independently of any earlier computation by five Python scripts that use only SymPy [10] (version 1.14.0 on Python 3.12.14) and exact arithmetic: rational numbers, polynomial identities in the symbols q and m , symbolic identities in t , real-root counting by Sturm sequences (SymPy’s `Poly.count_roots`) and real-root isolation by the Vincent-Akritas-Strzeboński continued-fraction method as implemented in SymPy’s `Poly.intervals` [1] (see [5, Chapters 2 and 10] for Sturm sequences and Descartes-based isolation in general), with limits computed symbolically. Both routines count distinct real roots; where a sign change is claimed it is witnessed by exact values at rational points. No floating-point comparison is used for any certified claim; decimal values quoted above are printed from isolating intervals of width below 10^{-12} . Each script prints one line per check and exits with status 0 only if every check passed; all 110 checks pass. The scripts are supplied as supplementary material in the directory `verify/` together with a runner `run_all.sh`.

`verify_core.py` (42 checks, about 25 s). Theorems 3.1 and 4.1: majorization, the identities (2), (3) and (6), the exact values, the irreducibility of B_1 (directly and by the reduction modulo 2), the root counts and isolating intervals for $m = 1, \dots, 64$ with the localisation $\frac{1}{2} - \frac{3}{16m} < q_*(m) < \frac{1}{2}$, Remark 4.2, Remark 5.3, variances and limits.
SHA-256 4660ba2f26025e26f6a68fc8375a6bbb44306bd258d4cc5a203e653a28818b74

`verify_general_mechanism.py` (16 checks, about 70 s). Theorem 5.1 and Corollary 5.2: the identities for R and F , the two derivatives, the limits, the case $w = 0$ and the specialisation to (4).
SHA-256 dd61c5493c737bc9baafc094684208e5e83448f4466d6211d70c119a8c4f7f68

`verify_strict_variant.py` (21 checks, about 6 s). Proposition 6.1: antiordering, the T -transform, the eight rational values, the factorizations with P and $\tilde{P}$, the coefficient sequences of Q and $\tilde{Q}$ used with Descartes’ rule, the Sturm counts and the survival difference.
SHA-256 fe993d1b1df7d7d7fce6e77d57d3635ede71235b41862527e298b1a10c3055d3

`verify_two_crossings.py` (13 checks, about 25 s). Proposition 6.2: majorization, the degree bookkeeping (279 before and 275 after cancelling $(1 + s + s^2)^2$), exact isolation of the two roots of the reduced numerator, signs at rational points, the survival difference, limits and variances.
SHA-256 06b4eb6b6c8e02393c5a548911ee5c7b75798b3c1b963276f434cb865da1414c

`verify_reversed_hazard.py` (18 checks, about 5 s). Section 7.3: the reduction (12) and the identity (13) for three components with symbolic baseline, weights, rates, θ and α ; the values $\frac{123}{1001}$, $\frac{11}{7}$ and $\frac{11}{25}$; the reversed hazard rates of the model (1.3) for instance (1) and for the pair (3, 5), (4, 4); and the $\alpha < 0$ computation.
SHA-256 a25d901588f1efbc6a3b4431384b985221bcf775d414b8511a045152074f064d

The audit layer. The claims added in Section 7 (the $\mathcal{W}_n$, $\alpha \leq 0$ instances (5) and (6), the Theorem 3.4 counterexample, the two invalid printed counterexamples, the bounded-search statistics, and the [16] errata) come from a separate audit of all finitely testable ordering claims in the two manuscripts, recorded in `memo.md` in the directory `mixture-audit` of the accompanying research repository and implemented by scripts that share the same exact-arithmetic discipline (exact rationals, Sturm counts and isolating intervals through SymPy; 60-digit evaluation where genuinely algebraic quantities are compared):

`memo.md`. The audit record: per-claim status, parameter tuples, witnesses and sample counts.
SHA-256 7af6759e843e22ec2cb558a0460ac90406211c5cb0ef89d02e5f066139b775ee

`certify_thm34.py`. The instance of the Theorem 3.4 paragraph: its hypotheses, the identity (11), the root counts and the witnesses 305/3888 and $-1579547560/2598836427243$, and the three-component instance mentioned there.
SHA-256 9bde7bca907e7ce679ac49232a5945dfd855db8d6fda982db4e8f7677d9fe6b0

`certify_whalf.py`. Instances (5) and (6): the $\mathcal{W}_3$ membership tests including the intermediate matrix of (6), the T -transforms, the identity (10) and the signed witnesses.
SHA-256 7734bae9c52d33053f82689252edad8b36ace72f4619f77cbad59facaaf3dd3b

`verify_paper_cex.py`. Rechecks the printed examples and counterexamples of both manuscripts, including the two invalid ones discussed above.
SHA-256 01812f258d539495decaa032abe4d614a43a764bf9de6a389cbd1cb1baf104e8

`audit_skf_st.py`, `audit_skf_hr.py`, `audit_saf.py`, with shared helpers `audit_lib.py`. The bounded searches behind the sample counts quoted above (stochastic order, hazard and reversed hazard claims of [14], and the tested claims of [16]); hypotheses are enforced exactly (the $\mathcal{V}_n/\mathcal{W}_n$ and $\mathcal{U}_n$ membership tests, majorization, weak sub- and supermajorization, and the T -transform factorizations).

Sample counts quoted in Section 7 are bounded-search statistics, not proofs.

References

- [1] Alkiviadis G. Akritas and Adam W. Strzeboński. A comparative study of two real root isolation methods. *Nonlinear Analysis: Modelling and Control*, 10(4):297–304, 2005. doi: 10.15388/NA.2005.10.4.15110.
- [2] Majid Asadi, Nader Ebrahimi, and Ehsan S. Soofi. The alpha-mixture of survival functions. *Journal of Applied Probability*, 56(4):1151–1167, 2019. doi: 10.1017/jpr.2019.72.
- [3] Ghobad Barmalzan, Sajad Kosari, and Narayanaswamy Balakrishnan. Orderings of finite mixture models with location-scale distributed components. *Probability in the Engineering and Informational Sciences*, 36(2):461–481, 2022. doi: 10.1017/S0269964820000467.
- [4] Jarosław Bartoszewicz and Magdalena Skolimowska. Preservation of stochastic orders under mixtures of exponential distributions. *Probability in the Engineering and Informational Sciences*, 20(4):655–666, 2006. doi: 10.1017/S0269964806060402.
- [5] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, Berlin, Heidelberg, second edition, 2006. doi: 10.1007/3-540-33099-2.
- [6] Raju Bhakta, Suchandan Kayal, and Narayanaswamy Balakrishnan. Ordering results between two multiple-outlier finite δ -mixtures. *Statistics & Probability Letters*, 213:110193, 2024. doi: 10.1016/j.spl.2024.110193.
- [7] Lina Guo and Rongfang Yan. Orderings of the finite mixture with modified proportional hazard rate model, 2024. URL <https://arxiv.org/abs/2407.15638>. arXiv:2407.15638v2 [math.ST], 23 July 2024.
- [8] Nil Kamal Hazra and Maxim Finkelstein. On stochastic comparisons of finite mixtures for some semiparametric families of distributions. *TEST*, 27(4):988–1006, 2018. doi: 10.1007/s11749-018-0581-7.
- [9] Albert W. Marshall, Ingram Olkin, and Barry C. Arnold. *Inequalities: Theory of Majorization and Its Applications*. Springer Series in Statistics. Springer, New York, second edition, 2011. doi: 10.1007/978-0-387-68276-1.
- [10] Aaron Meurer, Christopher P. Smith, Mateusz Paprocki, Ondřej Čertík, Sergey B. Kirpichev, Matthew Rocklin, AMiT Kumar, Sergiu Ivanov, Jason K. Moore, Sartaj Singh, Thilina Rathnayake, Sean Vig, Brian E. Granger, Richard P. Muller, Francesco Bonazzi, Harsh Gupta, Shivam Vats, Fredrik Johansson, Fabian Pedregosa, Matthew J. Curry, Andy R. Terrel, Štěpán Roučka, Ashutosh Saboo, Isuru Fernando, Sumith Kulal, Robert Cimrman, and Anthony Scopatz. SymPy: symbolic computing in Python. *PeerJ Computer Science*, 3:e103, 2017. doi: 10.7717/peerj-cs.103.
- [11] Hossein Nadeb and Hamzeh Torabi. New results on stochastic comparisons of finite mixtures for some families of distributions. *Communications in Statistics - Theory and Methods*, 51(10):3104–3119, 2022. doi: 10.1080/03610926.2020.1788082. Published online 9 July 2020.
- [12] Frank Proschan. Theoretical explanation of observed decreasing failure rate. *Technometrics*, 5(3):375–383, 1963. doi: 10.1080/00401706.1963.10490105.
- [13] Supriya Sahoo, Suchandan Kayal, and Narayanaswamy Balakrishnan. Ordering results between two finite α -mixture models with components following modified proportional hazard rate model. *Mathematics*, 14(14):2557, 2026. doi: 10.3390/math14142557.
- [14] Supriya Sahoo, Suchandan Kayal, and Maxim Finkelstein. Stochastic ordering results between two finite α -mixture models with resilience-scaled components. *Applied Stochastic Models in Business and Industry*, 42(2):e70089, 2026. doi: 10.1002/asmb.70089. Statement numbers and wording cited in this note refer to the peer-reviewed accepted author manuscript deposited in the University of Strathclyde repository (Strathprints record 96227, 26 pages).
- [15] Moshe Shaked and J. George Shanthikumar. *Stochastic Orders*. Springer Series in Statistics. Springer, New York, 2007. doi: 10.1007/978-0-387-34675-5.
- [16] Omid Shojaee, Majid Asadi, and Maxim Finkelstein. Stochastic properties of generalized finite α -mixtures. *Probability in the Engineering and Informational Sciences*, 36(4):1055–1079, 2022. doi: 10.1017/S0269964821000243. Published online 15 July 2021.